



Society of Petroleum Engineers

SPE-227400-MS

Coupled Lab-Scale and Numerical Analysis for Fault Critical Re-Active Stress Evaluation

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227400-MS](https://doi.org/10.2118/227400-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

Dual-horizontal-well SAGD, proven effective in Canada's oil sands and China's Fengcheng extra-heavy oil reservoir, faces challenges in shallow, high-viscosity reservoirs with strong heterogeneity—including thin muddy laminae and interbedded mudstones—resulting in prolonged preheating and uneven steam chamber development. This study integrates lab-derived rock mechanics data to establish a coupled stress field model, analyzing dilation characteristics during SAGD startup via geomechanical simulation. For shallow uncemented faults (common in high-porosity, unconsolidated formations), conventional fault-seal assessment methods (SGR/displacement pressure) prove inadequate. Applying Byerlee's slip criterion, we calculate critical fault instability pressures (2.74–4.26 MPa), recommending a reduced friction coefficient (0.4–0.5 versus typical 0.6–1.0) for water-saturated, unconsolidated fill zones. Reservoir dilation significantly enhances interwell properties, accelerating steam injection and establishing hydraulic connectivity to shorten preheating.

Technical Background

Extra-heavy oil/bitumen behaves as a solid in situ, with sand grains forming interconnected frameworks through geological compaction. Cemented in an interlocked texture, these grains undergo overriding, shearing, rotation, displacement, and deformation under confining stress. Shear-induced dilation at low confinement creates micro-fractures and high-permeability zones. Hydraulic fractures propagate perpendicular to the minimum principal stress; where the minimum principal stress is horizontal, vertical fractures form-enhancing SAGD's gravity drainage. Rapid startup technology injects high-pressure water into dual horizontal wells, inducing shear and tensile dilation to form near-well enhanced permeability zones that increase porosity, permeability, and mobile water saturation. Prior studies confirm dilation improves interwell properties, accelerates steam injection, and establishes hydraulic connectivity to shorten preheating. However, existing homogeneous models simulate dilation using finite-element software (e.g.,

ABAQUS) coupled with thermal simulators (e.g., CMG), neglecting ongoing geomechanical changes during production-limiting parameter optimization in heterogeneous reservoirs.

As shallow, thin, heavy-oil reservoirs are increasingly developed via intensive injection/production, polymer flooding, and steam injection-including offshore fields with uncertain geology-new challenges emerge, notably fault-seal integrity changes causing well failures and environmental risks. Batch casing damage in mature fields results from bedding-plane slippage and fault reactivation, evidenced by reservoir pressure depletion at Wilmington and surface fluid leaks at Jingtou. While lateral/vertical fault-seal assessment (via shale gouge ratio and displacement pressure) is well-established, current methods overlook stress alterations induced by injection/production. Additionally, experimental apparatus simulating non-condensable gas-assisted SAGD effectiveness was developed. The opening and sealing of faults play a decisive role in the process of hydrocarbon migration and accumulation. The evaluation of fault seal integrity is not only crucial for the study of hydrocarbon accumulation mechanisms and enrichment patterns, but it is also an essential part of the geological assessment before converting oil reservoirs into gas storages. Currently, the evaluation of fault seal integrity in hydrocarbon reservoirs primarily employs a combination of qualitative and quantitative methods, focusing on vertical and lateral sealing aspects. Vertical seal integrity of faults is mainly influenced by the degree of tightness of the fault plane; if the fault plane is tightly closed, the vertical seal integrity is good, preventing vertical hydrocarbon migration. Evaluation methods include normal stress on the fault plane and fault-reservoir differential pressure methods. Lateral seal integrity is primarily controlled by the lithology of the fault walls, with various evaluation methods such as the Allam diagram, distribution of fluid properties across the fault walls, juxtaposition of lithologies, shale gouge ratio, fault opening coefficient, and differential pressure methods.

In conventional dual-well SAGD (Fig. 1), horizontal/multiple vertical injectors above the producer deliver steam-gas mixtures, forming a chamber where gas accumulates near the interface due to condensation. Higher gas concentrations at the top and interface reduce chamber temperatures through vapor partial pressure depression. Three distinct thermal zones exist: only regions near injectors/producers reach steam temperature, while gas accumulation cools the upper chamber-lowering heat loss to overburden and reducing thermal requirements. For the injector-producer-separated configuration (Fig. 2), horizontal injectors (replaceable by vertical wells) maintain pressure via gas injection. Steam injection and production occur in the same well, with injected gas displacing oil toward producers. Near producers, reduced oil relative mobility requires operational parameter optimization to mitigate gas coning.

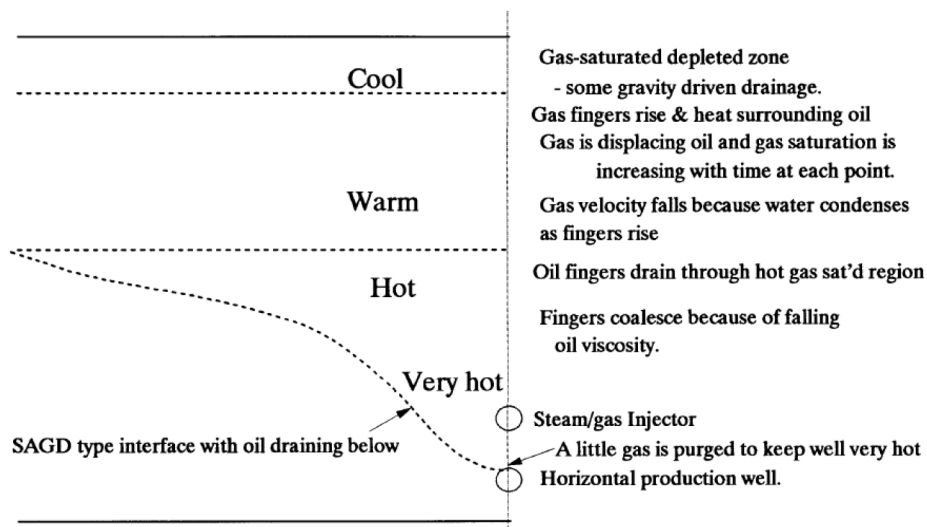


Figure 1—Conventional Dual-Well SAGD Configuration

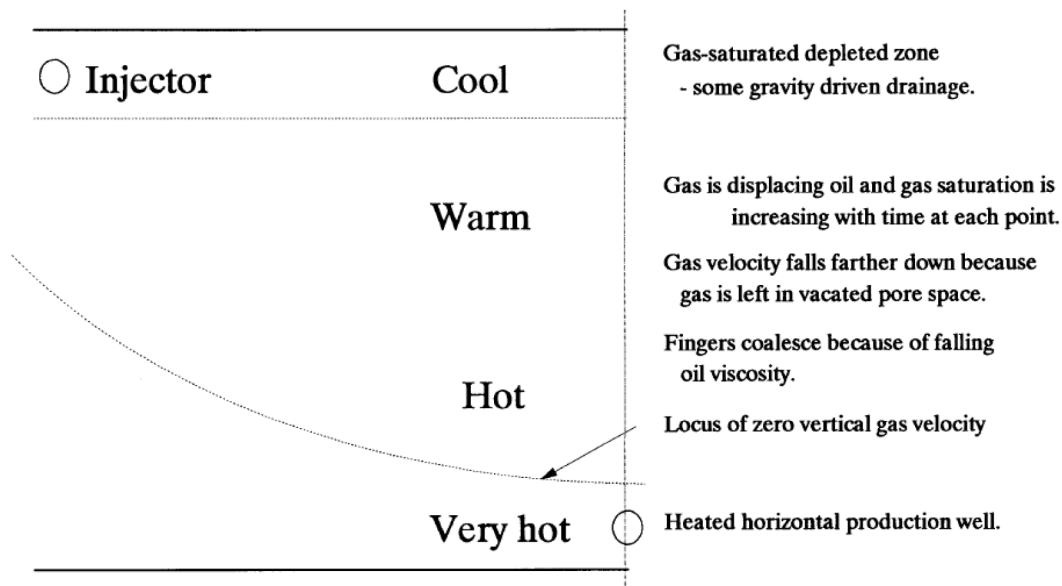


Figure 2—Conventional Dual-Well SAGD Process

Temperature within the steam chamber varies due to non-condensable gas accumulation, creating viscosity gradients along the interface. Higher oil viscosity at the upper interface requires steeper slopes to maintain potential gradients for flow. Interface geometry depends on thermal gradients: appropriate gradients enable vertical interfaces (Fig. 3), while excessive gradients cause instability in lower sections.

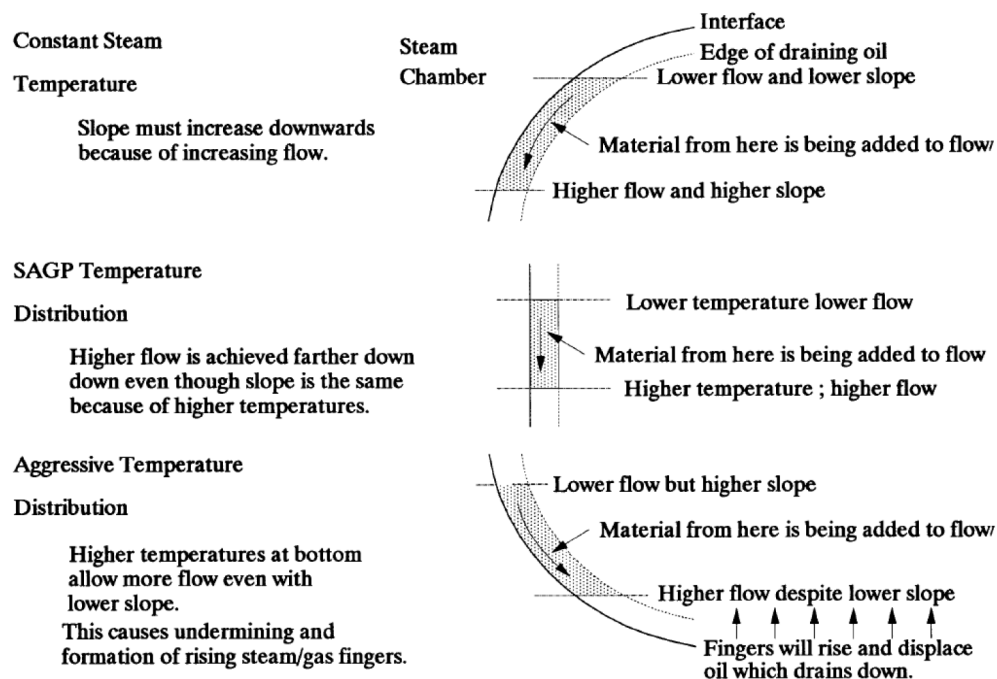


Figure 3—Potential Interface Morphologies Under Varied Thermal Gradients

In SAGD operations with constant chamber temperature, accumulating oil/condensate along the interface increases required pressure gradients. The potential gradient-proportional to interface slope-must drive larger fluid volumes toward producers, causing progressive steepening near the wellbore.

Fault Critical Failure Stress Calculation

The study area lacks sufficient capillary pressure data, constraining conventional SGR and fault-rock displacement pressure methods. To enable multi-faceted assessment of key fault sealing capacity, this study preliminarily explores rock mechanical approaches. Recent interpretations indicate the dominant fault system within the main oil-bearing area has lengths of 0.1-4.5 km and throws of 5-300 m. These extensional faults exhibit an en echelon pattern, with near-vertical upper sections transitioning to slightly listric geometries downward. Maximum vertical throws range from 20-75 m: predominantly 20-40 m in shallow and deep intervals, and 25-75 m in intermediate intervals. Peak throw frequencies occur within 20-40 m. Fault lengths vary by depth but cluster around 1-2 km, peaking at 1 km in intermediate and deep sections. Normal fault dips range from 40-80°, predominantly 55-60°, with shallower intervals (60-65°) generally exhibiting steeper dips than deeper sections (50-65°).

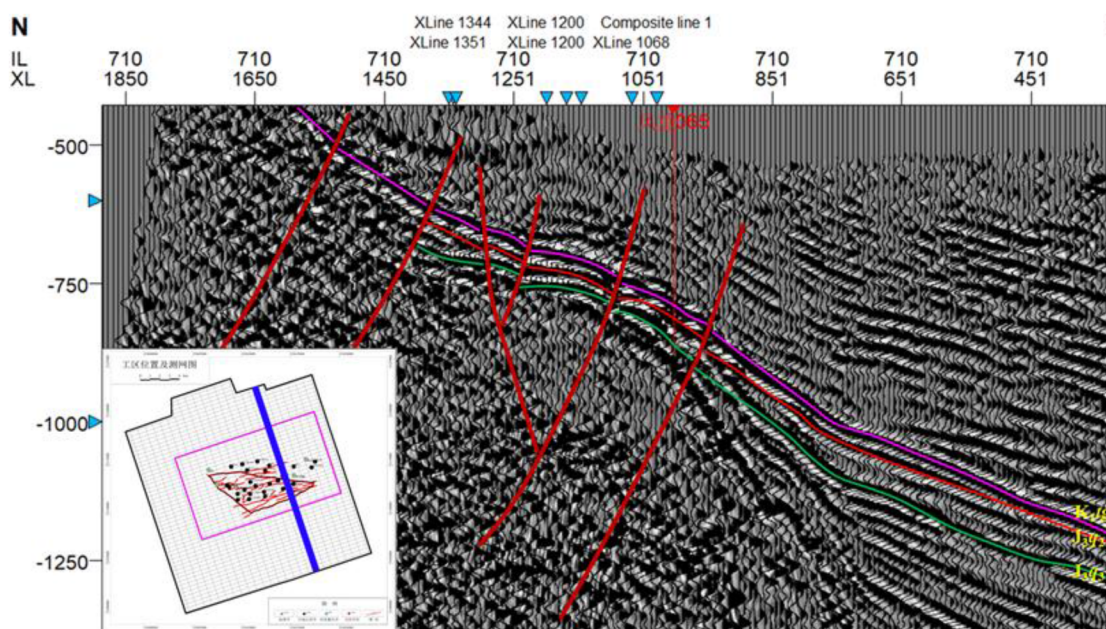


Figure 4—Typical Structural Section of Study Area

The critical slip pressure for a major fault in the study area was calculated using Byerlee's sliding criterion. Given the uncertainty in determining the appropriate friction coefficient, values ranging from 0.3 to 0.7 were assumed (Table 1, CFP:Critical Formation Pressure/MPa, FFP:Critical Fault Failure Pressure/MPa).

Table 1—Friction Coefficient Calibration Statistics

section	friction coefficient:0.7		friction coefficient:0.6		friction coefficient:0.5		friction coefficient:0.4		friction coefficient:0.3	
	CFP	FFP	CFP	FFP	CFP	FFP	CFP	FFP	CFP	FFP
S1	25.94	6.38	24.99	5.55	24.08	4.83	22.71	3.75	20.42	1.95
S2	25.71	5.54	24.62	4.46	23.57	3.59	21.99	2.27	19.38	0.08
S3	27.59	6.75	26.58	5.85	25.61	5.07	24.15	3.90	21.74	1.94
S4	27.38	6.85	26.27	5.91	25.20	5.10	23.61	3.89	20.96	1.85
S5	28.17	6.94	27.08	5.99	26.03	5.16	24.47	3.92	21.86	1.86
S6	29.06	7.24	28.05	6.33	27.08	5.54	25.61	4.32	23.16	2.33
S7	26.64	5.30	25.25	3.75	23.93	2.66	21.96	1.04	18.68	-1.67
S8	28.39	6.27	27.12	5.01	25.91	4.01	24.08	2.51	21.02	0.00
S9	27.86	5.70	26.40	4.13	25.02	2.99	22.97	1.31	19.55	-1.50
S10	29.55	6.42	28.26	5.07	27.03	4.01	25.17	2.39	22.07	-0.29
S11	31.07	7.99	29.85	6.92	28.68	5.99	26.93	4.61	24.02	2.28
S12	31.52	7.42	30.24	6.18	29.01	5.15	27.17	3.62	24.11	1.04
S13	28.87	5.72	27.18	3.66	25.59	2.34	23.19	0.38	19.20	-2.91
S14	31.15	7.01	29.69	5.58	28.29	4.44	26.21	2.73	22.73	-0.12
S15	32.66	7.33	31.05	5.72	29.52	4.46	27.24	2.60	23.43	-0.53
S16	32.56	6.35	30.90	4.19	29.33	2.76	26.96	0.60	23.00	-2.99

Friction Coefficient Determination Methodology

Field observations indicate that gas injection wells near faults experience a sharp injection rate increase and pressure drop at 5.25 MPa. Pressure recovery occurs rapidly after shutting in or even backflowing water injectors. This suggests that 5.25 MPa slightly exceeds the critical additional fluid pressure for fault stability; exceeding this downhole pressure triggers fault instability. Although Byerlee's sliding criterion typically employs friction coefficients (FC) of 0.6-1.0 in industrial applications, even its lowest recommended value exceeds the stability limit for faults in this area. Therefore, the applicability and appropriate FC range for Byerlee's criterion were investigated. Research indicates that fault cores form at throws exceeding 5 m, likely present in this area. This raises questions about the suitability of Byerlee's FC range, derived from block-on-hard-surface tests, for these faults. Key experimental modifications included: 1) applying a 1-mm-thick gouge layer between rock samples, and 2) considering water saturation effects. Results show: 1) frictional properties are condition-dependent, and 2) FC values range from 0.2-0.8, with lower values (0.2-0.5) observed at small displacements (<0.5 mm). The lower FCs are attributed to the lubricating effect of the gouge, contrasting with Byerlee's rigid surface assumption. Given the study area's faults contain water-saturated, poorly consolidated gouge, similar to the experimental setup, a maximum Byerlee FC of 0.6 is recommended. Combined with sensitivity analysis, a FC range of 0.4-0.5 is proposed for the area.

Normal Stress on the Fault Plane. The vertical sealing capacity of a fault is primarily influenced by the degree of closure of the fault plane. If the fault plane is tightly closed, the fault exhibits good vertical sealing capacity, preventing hydrocarbons from migrating vertically along the plane. Conversely, if the fault is open, it can serve as a conduit for hydrocarbon migration. The degree of closure of the fault plane can be measured by the magnitude of the normal stress on the fault plane, which is calculated using the following formula:

$$P = H(\rho_r + \rho_w) \times 0.009876 \cos\theta \quad (1)$$

P is the normal stress on the fault plane, in MPa; H is the burial depth of the fault plane, in meters;

ρ_r is the average density of the overlying strata, in g/cm³; ρ_w is the density of the formation water, in g/cm³; θ is the dip angle of the fault plane, in degrees (°).

The current maximum horizontal principal stress direction in the YH block is predominantly near east-west. The normal stress on the fault planes of various faults in the YH block was calculated using Equation 1, and the results are shown in Table 2. The results indicate that the normal stress on the fault planes in the YH block ranges from 8.48 to 47.43 MPa, which is significantly greater than 7.5 MPa. Combined with SGR values generally exceeding 40% and corresponding sand-to-shale ratios less than 60%, according to the evaluation standards for vertical fault sealing capacity, the vertical sealing capacity of the faults falls within the medium to good range.

Table 2—Sectional positive pressure evaluation of fault section in the YH Block

Fault type	Fault No.	SGR	Fault Dip	Positive Pressure/MPa	Fault sealing
Class I Fault	F1	0.62	50	41.84	Good
	F2	0.60	53	21.10	Good
	F3	0.50	65	11.59	Good
	F4	0.51	51	17.34	Good
	F5	0.52	75	8.48	Good
	F6	0.44	68	25.32	Better
Class II Fault	F7	0.67	72	47.43	Good
	F8	0.75	52	18.34	Good
Class III Fault	F9	0.72	53	36.99	Good
	F10	0.78	53	36.56	Good
	F11	0.38	50	60.61	medium

Specifically, the SGR of Class I faults ranges from 44% to 62%, with corresponding sand-to-shale ratios between 38% and 56%, generally less than 50%. The normal stress on the fault planes ranges from 8.48 to 41.84 MPa, significantly greater than 7.5 MPa, indicating good vertical sealing capacity for these faults. Only the F6 fault has a relatively high sand-to-shale ratio, yet it still exhibits good vertical sealing capacity. Class II faults have SGR values greater than 60% and corresponding sand-to-shale ratios less than 40%, with normal stress on the fault planes significantly greater than 7.5 MPa, indicating good vertical sealing capacity. The vertical sealing capacity of Class III faults ranges from medium to good. For instance, the F11 fault has a lower SGR and a sand-to-shale ratio of 62%. Despite the relatively high normal stress on the fault plane, the higher sand-to-shale ratio suggests that its vertical sealing capacity is medium.

Lateral Sealing Capacity of Faults

Fault Opening Factor Analysis. The fault opening factor is also commonly used to evaluate the lateral sealing capacity of faults. The calculation formula is:

$$F_{OI} = \frac{P_f}{\delta \cdot SGR} \quad (2)$$

$$\delta = (\sin \alpha \cdot \sin \theta)^2 \sigma_H + (\cos \alpha \cdot \cos \theta)^2 \sigma_h + \cos^2 \theta \sigma_v \quad (3)$$

FOI is the fault opening index, ranging from 0.5 to 5.0; the larger the value, the poorer the fault sealing capacity. δ is the normal stress on the fault plane, in MPa. α is the angle between the fault strike and the direction of the maximum horizontal principal stress, in degrees ($^{\circ}$). σ_H , σ_h , and σ_v are the maximum horizontal principal stress, minimum horizontal principal stress, and vertical stress, respectively, in MPa. P_f is the fluid pressure, in MPa.

Dynamic Sealing Capacity of Faults. Faults are important barriers for the safety of gas storage reservoirs. In oil and gas reservoirs that are being converted into gas storage facilities, boundary faults are generally sealed in their undeveloped original state. However, during the intensive injection and extraction processes in gas storage operations, faults may become reactivated, open, or experience slip instability. During the injection and extraction processes in gas storage, the stability of faults may be threatened. Fault stability refers to the tendency for fault reactivation, which can increase with the amount of injection in the reservoir adjacent to the fault. As pressure gradually increases, the characteristics of the stress field borne by the fault change, and the effective stress gradually decreases, approaching the critical value for fault reactivation pressure. The reactivation pressure (PR) is the total fluid pressure required for fault reactivation under the initial in-situ stress conditions, which is the sum of the hydrostatic pressure at a certain depth and the additional fluid pressure needed for fault reactivation. The calculation formula is:

$$P_R = \rho_w g H + \Delta P. \quad (4)$$

P_R is the fault reactivation pressure, in MPa. ΔP is the additional fluid pressure, in MPa. Due to the lack of fault rock samples, samples from the D segment are used as substitutes to simulate and calculate the critical pressure for fault reactivation and opening.

Fault Slip Instability. Geomechanical studies indicate that during the stress disturbances caused by gas storage injection and extraction, a fault will undergo slip instability and lose its sealing ability when the shear stress acting on the fault plane exceeds the product of the friction coefficient and the effective normal stress. Currently, the fault slip index is commonly used to evaluate fault stability under the disturbances of gas storage operations, defined as the ratio of shear stress to effective normal stress on the fault plane. When this ratio is less than 0.6, the fault is considered to have good mechanical stability. A ratio of 0.6 or higher indicates a risk of slip instability, with higher ratios corresponding to greater risk. The Analytic Hierarchy Process (AHP) is a decision-making analysis method that combines qualitative and quantitative approaches, and it is an effective method for determining weights. The basic principle of AHP is to decompose complex problems into several influencing factors (or evaluation indicators) and organize these factors into an orderly hierarchical structure based on their dominance relationships. By making pairwise comparisons, the relative importance of the factors within each level is determined, which then helps in deciding the order of relative importance of all factors. It clarifies fuzzy concepts to determine the importance order of all factors.

Heavy Oil Dilation Characteristics

This method enhances conventional cyclic steam stimulation by introducing convective inter-well heat transfer, accelerating fluid/thermal connectivity and reducing preheating time. The rapid-startup sequence comprises: 1) Stress adjustment: Producer injection slightly above initial reservoir pressure modifies near-wellbore stress and rock properties. 2) Fracture-initiated dilation: Simultaneous injection through both injector and producer wells above minimum principal stress (injector rate > producer) synchronizes dilation, establishing inter-well connectivity with increasing near-well porosity. 3) Dilation propagation: Continued injection extends dilation zones, improving inter-well reservoir quality while reducing oil saturation and mobilizing bound water.

Rock mechanics parameters are critical for reservoir dilation design. Triaxial compression tests on standardized cores from representative wells determined key properties (friction angle, cohesion, dilation angle) for oil sands, caprock, and interbeds under reservoir-relevant confining pressures (4-6 MPa). Caprock exhibits significantly higher cohesion (4.6 MPa vs. 0.9 MPa for oil sands), yet comparable shear strengths create stability risks. Shear/tensile fractures propagating upward from dilated oil sands could compromise caprock integrity due to insufficient sealing capacity, necessitating operational pressures below caprock fracture pressure. Constitutive models derived from laboratory stress-strain curves provide the theoretical basis for dilation design.

Oil sands exhibit increased volumetric dilation with axial strain under low effective confining pressure, whereas elevated confining pressures reduce dilation rate and magnitude, demonstrated by dilation angles measuring 46 degrees at 0 MPa, 23 degrees at 2 MPa, 20 degrees at 3 MPa, 12 degrees at 4 MPa, and 9 degrees at 6 MPa. Given that dilation capacity typically decreases with higher confinement, effective reservoir dilation requires continuous gas injection below formation fracture pressure to increase near-wellbore pore pressure and reduce effective confinement, with injection pressure progressively elevated for deeper dilation targets. Mini-fracture testing provides reliable in-situ stress measurements by generating tensile fractures through limited fluid injection; subsequent pressure decline analysis determines fracture closure pressure, equivalent to the minimum horizontal principal stress. Stress gradients in Fengcheng Field's Wellblock 18 are: oil sands reservoir - vertical stress gradient 22.0 kilopascals per meter, maximum horizontal principal stress gradient 18.0 kilopascals per meter, minimum horizontal principal stress gradient 21.0 kilopascals per meter; caprock - vertical stress gradient 21.2 kilopascals per meter, maximum horizontal

principal stress gradient 28.0 kilopascals per meter, minimum horizontal principal stress gradient 21.0 kilopascals per meter (caprock fracture pressure approximately 10 megapascals). Stress orientation is interpreted from drilling-induced wellbore failures.

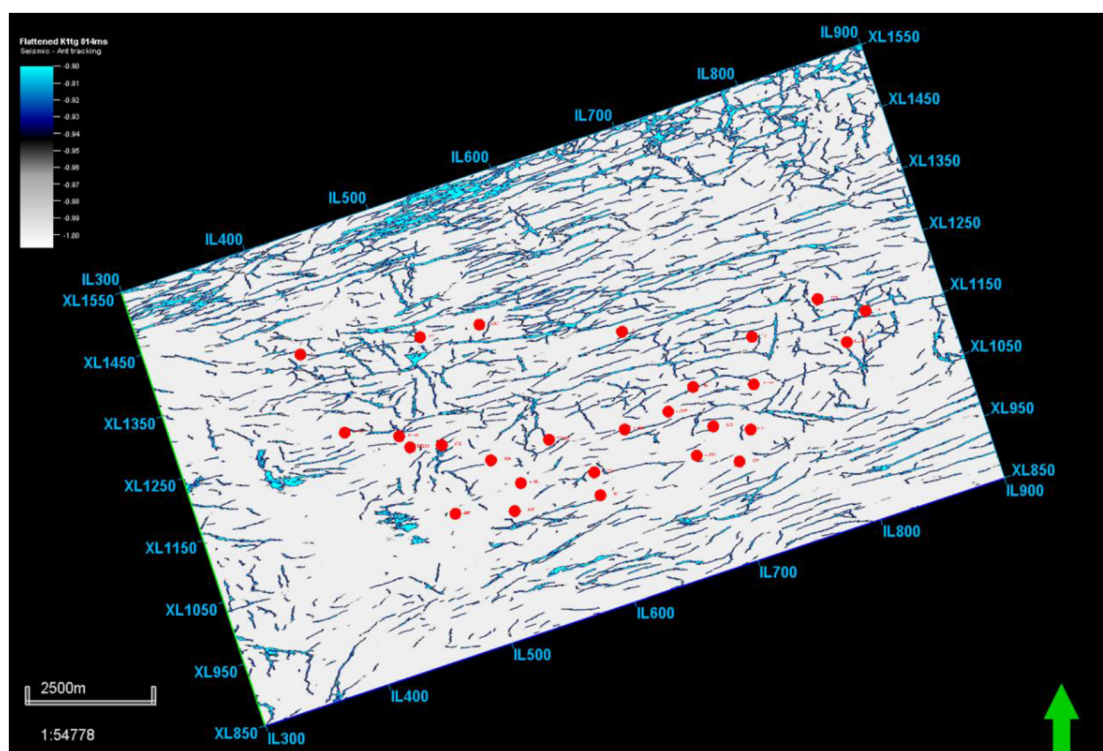


Figure 5—Natural Fracture Prediction in Study Area

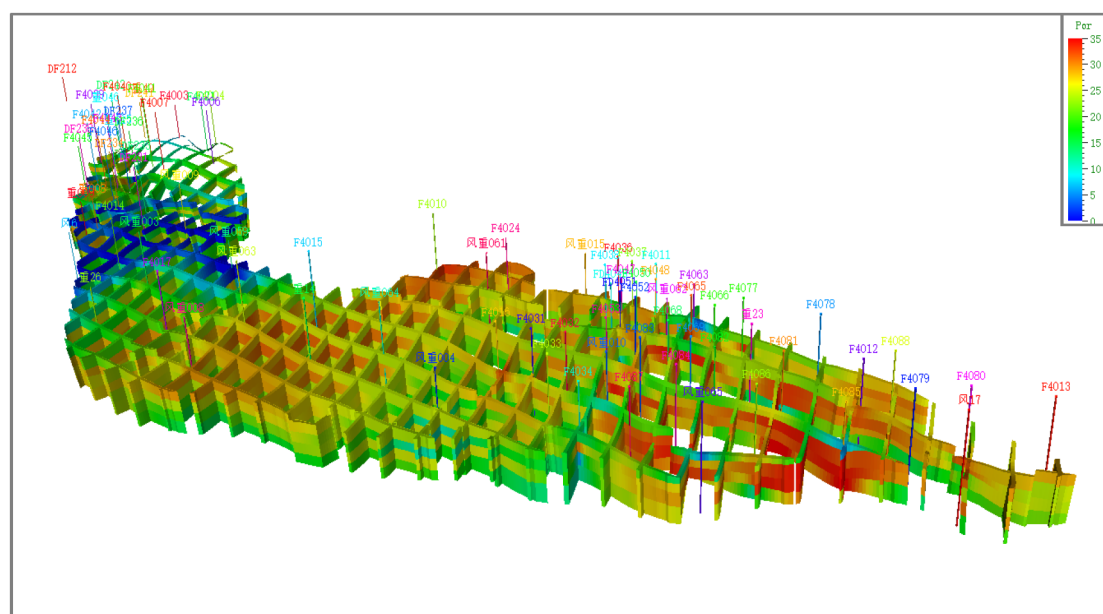


Figure 6—3D Geologic Modeling of Study Area

Wellbore collapse azimuth aligns with the minimum horizontal principal stress direction, while induced tensile fractures trend parallel to the maximum horizontal principal stress. Image log data from Fengcheng Field indicate a maximum horizontal principal stress orientation of N60°E. Triaxial strength tests

calibrated the oil sand strength model, demonstrating strong agreement between numerical simulations and experimental data. This rock mechanics calibration established constitutive relationships, enabling history-matched simulations of mini-fracture tests to characterize nonlinear permeability and fluid-solid coupling parameters for dilation modeling. Sensitivity analysis of mechanical parameters revealed Young's modulus as the dominant control on total dilation injection volume - injection decreases proportionally with increasing modulus. Other parameters (Poisson's ratio, cohesion, friction angle) exhibit weaker correlations with injection volume, primarily influencing minimum dilation pressure rather than total injection. Reservoir dilation primarily occurs through grain sliding/separation (not fracturing) due to high initial porosity, with Young's modulus (governing axial elasticity) exerting greater impact than Poisson's ratio (governing lateral deformation) on injection requirements. Compared to previously proposed evaluation methods, the fault sealing capacity evaluation index proposed in this paper has advantages in two aspects: Firstly, it considers a more comprehensive range of factors. It not only utilizes common indicators from oil and gas reservoir sealing evaluation methods but also considers factors affecting fault stability, which must be considered when converting oil and gas reservoirs into gas storage facilities. By evaluating fault sealing capacity from both static and dynamic dimensions, the results are more reliable and provide important guidance for the geological evaluation of oil and gas reservoir-type gas storage facilities. Secondly, the evaluation method is more comprehensive. The process of oil and gas accumulation is complex, and merely evaluating the vertical or lateral sealing capacity of faults makes it difficult to comprehensively reflect the effectiveness of accumulation. Statistical analysis shows that the correlation coefficient between a single factor and reservoir trap saturation is less than 0.1. However, this paper determines the weight coefficients for comprehensive evaluation based on the correlation between various evaluation indicators of controlling faults and trap fill level. The established comprehensive evaluation index has a good positive correlation with trap fill level (with a correlation coefficient of up to 0.645), allowing for a more accurate and comprehensive reflection of the effectiveness of oil and gas accumulation.

Conclusion

The SAGD rapid start-up technology enhances reservoir properties around the wellbore, increases thermal convection between wells, accelerates inter-well connectivity, and shortens the preheating cycle. This technology has been tested and widely applied in the Xinjiang Oilfield. Statistical results from wells with implemented SAGD rapid start-up show a reduction in the preheating cycle by 3.5 to 6.5 months and a decrease in steam consumption by 45.2% to 64.5%.

Analysis of fault critical failure pressure and field conditions indicates that the Byerlee slip criterion, based on the Coulomb-Mohr failure criterion, is suitable for analyzing fault instability in the oilfield. However, the friction coefficient range (0.6-1.0) derived from spring-slider experiments may not align with field conditions. Under water-saturated, unconsolidated fault gouge conditions, the friction coefficient range needs to be reassessed. Preliminary analysis suggests that a friction coefficient of 0.4-0.5 better reflects field conditions, leading to a determined critical fault failure pressure of 2.74-4.26 MPa for the study area.

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